POLITECNICO DI TORINO · Master Degree in Data Science

Official Examination Session 4

Date: September 16, 2026 | **Time Allotted:** 1 Hour 30 Minutes | **Total Points:** 16.00 Points

Session Scope
Autumn Examination Session

§1 Exam Session 4 — September 16, 2026

Session Overview: This examination comprises 7 questions covering inherently interpretable model criteria, mechanistic interpretability pitfalls and polysemanticity, surrogate modeling theory (global vs. local surrogates), Concept Bottleneck Models (CBMs) versus Testing with Concept Activation Vectors (TCAV), adversarial attack danger motivations (including the absence of effective defenses), clinical saliency map evaluation strategies (faithfulness vs. plausibility, AOPC, and sanity checks), and a dual-stakeholder tabular XAI design for student dropout prediction (total: 16.00 points).

Question 1: Inherent Interpretability and User Constraints — MCQ — 1.00 Point

Which of the following is the most accurate statement about inherently interpretable models?

a. They are always preferable to black-box models because they guarantee both high accuracy and explainability.

b. They are interpretable only if their learned structure and input features remain understandable to the intended users.

c. They are equivalent to post-hoc explanation methods because they explain a black-box model after training.

d. They are considered interpretable because they always produce explanations that are independent of the input features.

✓ Executive Solution Summary

The correct answer is **b**. An inherently interpretable model (e.g., decision trees, sparse linear models, rule lists) is not unconditionally interpretable in practice. It remains interpretable **only if its structural complexity is bounded** (e.g., shallow tree depth, small cognitive budget) and its **underlying input features carry semantic meaning** for human users.

➤ Systematic Analysis of Multiple Choice Options

(b) [CORRECT] They are interpretable only if their learned structure and input features remain understandable to the intended users.

Inherently interpretable models (interpretable-by-design) depend on two strict conditions for actual interpretability: (1) *simulatability and decomposability* (a human must be able to trace the computation, which fails if a decision tree has 2,000 leaves or a regression has 10,000 coefficients), and (2) *feature interpretability* (features must be human-understandable; a linear model trained on 100 abstract PCA components or raw

pixel embeddings is completely opaque).

(a) [INCORRECT] They are always preferable to black-box models because they guarantee both high accuracy and explainability.
False. Inherently interpretable models often suffer from an accuracy-interpretability trade-off, particularly on complex high-dimensional unstructured data (vision, audio, natural language) where deep neural architectures significantly outperform shallow linear models or trees.

(c) [INCORRECT] They are equivalent to post-hoc explanation methods because they explain a black-box model after training.
False. Inherently interpretable models are **by-design (in-modeling)** architectures whose internal mechanics are transparent during inference. Post-hoc methods explain an already trained black box externally.

(d) [INCORRECT] They are considered interpretable because they always produce explanations that are independent of the input features.
Nonsensical. Interpretability relies fundamentally on understanding how output decisions depend on input features. An explanation independent of inputs would be completely uninformative and unfaithful.

✍ **Comprehensive Technical Walkthrough**

In the taxonomy of XAI, **In-Modeling Interpretability** (interpretable-by-design) requires meeting strict cognitive and structural bounds (Lipton, 2018):

Criterion 1. Simulatability (Model-Level Transparency): The ability of a human to comprehend the model in its entirety and mentally compute the output in reasonable time. A decision tree satisfies this only if its depth $d \leq 4$ or number of leaves $|L| \leq 16$.

Criterion 2. Decomposability (Component-Level Transparency): Each input feature, parameter, and operational calculation must admit an intuitive explanation without requiring complex transformations.

Criterion 3. Semantic Feature Alignment: Even a trivial linear model $f(x) = \sum w_i x_i$ is uninterpretable if x_i represents abstract frequency transforms, latent PCA projections, or arbitrary polynomial combinations.

★ **Examiner Feedback & Common Pitfalls (Prof. Eliana Pastor)**

Students often fall into the trap of option (a) by asserting that interpretable models are "always better". Prof. Eliana Pastor emphasizes that interpretability is context-dependent and user-constrained: an overly complex decision tree is as opaque as a deep neural network.

➥ **Course Theory Mapping (Notes & Slides)**

NOTES/XAI_Theory.pdf – Chapter 4: "In-Modeling Explainability", Sections: "Goals and Scope", "Interpretability of Decision Trees", and "Linear Regression Limitations" (pp. 22–28).

¶ **Alternative Perspectives & Methodological Variants**

Generalized Additive Models (GAMs) with pairwise interactions (GA2Ms / Explainable

Boosting Machines) maintain exact inherent interpretability by constraining the function to $f(x) = \sum f_i(x_i) + \sum f_{ij}(x_i, x_j)$, preventing higher-order uninterpretable interactions.

Question 2: Mechanistic Interpretability Pitfalls and Concept Learning MCQ 1.00 Point

Which of the following is **incorrect** regarding mechanistic interpretability in deep learning?

a. A neuron is defined polysemantic when it fires for multiple patterns.

b. It is used to disentangle the hidden representation of a deep learning model (e.g. an LLM) which is otherwise a black-box.

c. When a feature is identified, we are sure the model has learnt the underlying concept.

d. It is a reverse-engineering process aimed to understand the decision-making process of a model.

✓ Executive Solution Summary

The correct answer is **c** (the incorrect statement). Identifying a feature direction or neuron that activates in the presence of an empirical pattern **does not guarantee** that the model has learned the true underlying causal concept. The activation may stem from spurious correlations, dataset shortcuts, or polysemantic superposition.

➤ Systematic Analysis of Multiple Choice Options

(c) [CORRECT] When a feature is identified, we are sure the model has learnt the underlying concept.

This statement is provably **INCORRECT**. In mechanistic interpretability, isolating an active direction or unit does not prove semantic concept acquisition: (1) *Correlation vs. Causality*: the unit may fire due to a co-occurring visual artifact or background shortcut, (2) *Superposition*: due to polysemanticity, a single direction can be an entangled linear projection of multiple unrelated concepts, and (3) *Functional Inaction*: ablating the feature might have zero causal impact on the downstream task.

(a) [INCORRECT] A neuron is defined polysemantic when it fires for multiple patterns.

This is a **TRUE** definition. A neuron is monosemantic if it responds to a single distinct concept, and polysemantic if it activates for multiple unrelated features.

(b) [INCORRECT] It is used to disentangle the hidden representation of a deep learning model (e.g. an LLM) which is otherwise a black-box.

This is a **TRUE** statement. Disentangling hidden states (e.g., using Sparse Autoencoders on LLM residual streams) is a core mission of mechanistic interpretability.

(d) [INCORRECT] It is a reverse-engineering process aimed to understand the decision-making process of a model.

This is a **TRUE** statement. Mechanistic interpretability explicitly seeks to reverse-engineer neural weights into computational circuits and algorithms.

✍ **Comprehensive Technical Walkthrough**

In mechanistic interpretability, establishing that a network has learned a concept requires more than identifying an activating feature $\mathbf{v} \in \mathbb{R}^d$:

- **The Polysemanticity / Superposition Trap:** Under the Superposition Hypothesis (Olah et al., 2020), neural networks pack M features into $D \ll M$ dimensions. An identified neuron or direction $\mathbf{v}$ is often an interference pattern of disparate semantic concepts:

$$\mathbf{h}(x) = \sum_{k=1}^{M} c_k(x)\mathbf{v}_k \implies \text{Neuron}_i = \mathbf{e}_i^T \mathbf{h}(x) = \sum_{k=1}^{M} c_k(x)(\mathbf{e}_i^T \mathbf{v}_k) \tag{1}$$

  Even if $\mathbf{e}_i$ fires whenever "Golden Retriever" appears, it may also fire for "yellow tennis ball" and "car wheels".

- **Causal Interventions vs. Observational Activations:** To confirm that a model has truly acquired an internal concept, researchers must perform **causal mediation analysis** or **activation patching**: intervening on $\mathbf{v}$ ($h \leftarrow h + \alpha\mathbf{v}$) must causally steer model behavior in the exact predicted manner.

★ **Examiner Feedback & Common Pitfalls (Prof. Eliana Pastor)**

Do not confuse observational correlation (a neuron lighting up on a test dataset) with causal concept learning. The exam question tests whether candidates recognize that empirical feature identification can be deceptive without causal ablation and circuit verification.

➥ **Course Theory Mapping (Notes & Slides)**

NOTES/XAI_Theory.pdf – Chapter 11: "Mechanistic Interpretability", Sections: "The Three Speculative Claims", "Features", and "Challenges: Polysemanticity and Superposition" (pp. 155–162).

¶ **Alternative Perspectives & Methodological Variants**

Sparse Autoencoders (SAEs) train an overcomplete sparse dictionary $\mathbf{h} \approx \sum f_i \mathbf{d}_i$ to disentangle polysemantic directions into monosemantic concept atoms, enabling true causal validation.

Question 3: Surrogate Models in XAI: Foundations, Global vs. Local, and Techniques

Open Inquiry | 1.50 Points

Outline the concept of surrogate models in XAI. Explain how surrogate models can be used to interpret the behavior of machine learning models. Distinguish between global and local surrogate models, and provide one example of a technique based on surrogate modeling.

✓ **Executive Solution Summary**

A **surrogate model** g is an inherently interpretable model (e.g., decision tree, linear regression, rule list) trained to approximate the predictions of an opaque black-box model f. Instead of training on true ground-truth labels y, the surrogate optimizes against the black-box outputs $\hat{y} = f(x)$, directly mimicking its decision boundary.
Global surrogates approximate the black box over the entire dataset/feature space $\mathcal{D}$, offering

high-level behavioral overviews but suffering from low fidelity on complex non-linear models. **Local surrogates** approximate the black box strictly within the neighborhood $\mathcal{N}(x)$ of a specific query instance, achieving high local fidelity. Canonical examples include **TREPAN** (global tree surrogate) and **LIME** (local linear surrogate).

✍ **Comprehensive Technical Walkthrough**

1. Conceptual Definition of Surrogate Models

Surrogate modeling is a **post-hoc, model-agnostic** explainability paradigm. Given a trained black-box model $f : \mathcal{X} \to \mathcal{Y}$, direct inspection of f is intractable. A surrogate model $g \in \mathcal{G}$ is chosen from a family of inherently interpretable models and trained to optimize **fidelity** (faithfulness to f):

$$\min_{g \in \mathcal{G}} \mathcal{L}(f, g) + \Omega(g) \tag{2}$$

where $\mathcal{L}$ measures the discrepancy between $f(x)$ and $g(x)$, and $\Omega(g)$ penalizes surrogate model complexity (e.g., tree depth, number of non-zero linear coefficients).

2. Global vs. Local Surrogate Models

Dimension	Global Surrogate Models	Local Surrogate Models
Scope	Global: Entire input manifold $\mathcal{X}$	Local: Specific neighborhood $\mathcal{N}(x)$ of query x
Optimization Objective	$\min_{g \in \mathcal{G}} \sum_{i=1}^{N} \mathcal{L}\left(f(x_i), g(x_i)\right) + \Omega(g)$	$\min_{g \in \mathcal{G}} \sum_{z} \pi_x(z) \mathcal{L}\left(f(z), g(z)\right) + \Omega(g)$
Core Assumption	A simple model can summarize global logic	A complex non-linear surface is locally linear
Key Trade-off	Global overview vs. poor fidelity on non-linearities	High local fidelity vs. zero global coverage
Canonical Example	**TREPAN**, Global Decision Trees	**LIME**, **LORE**

3. Concrete Technique Walkthrough: LIME (Local Surrogate)

LIME (Local Interpretable Model-agnostic Explanations) serves as the canonical local surrogate:

Step 1. Input Perturbation: Given an instance x and its binary interpretable representation $x' \in \{0, 1\}^M$, LIME generates a dataset of perturbed binary instances $Z' = \{z'_1, \ldots, z'_K\}$ by randomly sampling feature subsets.

Step 2. Black-Box Evaluation: The perturbed samples are mapped back to original feature space $h_x(z')$ and evaluated by the black box: $y_z = f(h_x(z'))$.

Step 3. Proximity Weighting: Each perturbation is weighted by an exponential distance kernel measuring proximity to x:

$$\pi_x(z) = \exp\left(-\frac{D(x, z)^2}{\sigma^2}\right) \tag{3}$$

Step 4. Surrogate Training: LIME fits a sparse linear model $g(z') = w_0 + \sum_{j=1}^{M} w_j z'_j$ minimizing weighted squared loss:

$$\mathcal{L}(f, g, \pi_x) = \sum_{z, z' \in Z} \pi_x(z) \left(f(z) - g(z')\right)^2 + \lambda \|w\|_0 \tag{4}$$

The resulting weights w_j represent local feature attributions for instance x.

★ Examiner Feedback & Common Pitfalls (Prof. Eliana Pastor)

When explaining surrogates, candidates often forget to state that **surrogate models are trained on the black box predictions** $f(x)$, NOT on the true dataset labels y. If a surrogate were trained on y, it would be an independent predictive model, not a surrogate explainer!

➥ Course Theory Mapping (Notes & Slides)

NOTES/XAI_Theory.pdf – Chapter 5: "Post-Modeling Explainability", Sections: "Global Surrogate Models" (pp. 31–34) and "Local Surrogate Models: LIME" (pp. 41–45).

¶ Alternative Perspectives & Methodological Variants

TREPAN is the classic global surrogate: it queries the black-box model using active sampling to reconstruct a decision tree using an information gain split criterion augmented with an m-of-n boolean operator.

Question 4: Concept Bottleneck Models (CBMs) vs. Testing with Concept Activation Vectors (TCAV)

Open Inquiry **3.00 Points**

Explain the difference between Concept Bottleneck Models (CBMs) and Testing with Concept Activation Vectors (TCAV). Define both approaches, discuss their main similarities and differences, and explain in which scenario you would prefer one method over the other.

✓ Executive Solution Summary

Concept Bottleneck Models (CBMs) are **explainable-by-design (in-modeling)** architectures that enforce a structural intermediate bottleneck layer $x \xrightarrow{g} c \xrightarrow{f} y$, mapping inputs to human concepts and concepts to task predictions. **TCAV (Testing with Concept Activation Vectors)** is a **post-hoc** explainability framework that calculates the directional derivative of a pre-trained network's class logit along a Concept Activation Vector ($\mathbf{v}_C^l$) learned in an intermediate activation space.
Both methods ground explanations in high-level human semantic concepts rather than pixels. However, they differ in timing (by-design vs. post-hoc), retraining requirements, faithfulness guarantees, and intervention capability. CBMs are preferred when building high-stakes systems requiring runtime human intervention; TCAV is preferred when auditing pre-trained black-box models where dense concept annotations do not exist.

✍ Comprehensive Technical Walkthrough

1. Formal Definitions

A. Concept Bottleneck Models (CBMs — In-Modeling / By-Design)

A CBM decomposes the machine learning pipeline into two explicitly constrained sub-networks:

$$x \xrightarrow{g} \hat{\mathbf{c}} \xrightarrow{f} \hat{y} \tag{5}$$

where $g : \mathcal{X} \to [0,1]^K$ is the *concept encoder* predicting K user-defined concepts (e.g., wing color, beak shape), and $f : [0,1]^K \to \mathcal{Y}$ is the *task predictor* (typically a sparse linear model or shallow tree).

- **Training Objective (Joint Training):**

$$\min_{\theta_g, \theta_f} \frac{1}{N} \sum_{i=1}^{N} \left(\mathcal{L}_{\text{task}} \left(f(g(x_i)), y_i \right) + \lambda \mathcal{L}_{\text{concept}} \left(g(x_i), c_i^* \right) \right) \tag{6}$$

- **Test-Time Intervention:** If a radiologist observes an erroneous concept prediction $\hat{c}_j$, they can overwrite it with the true value c_j^*. The task prediction automatically recalculates without retraining: $\hat{y}_{\text{new}} = f(c_1, \ldots, c_j^*, \ldots, c_K)$.

B. Testing with Concept Activation Vectors (TCAV — Post-Hoc)

TCAV (Kim et al., 2018) provides post-hoc concept attributions for an unmodified, pre-trained neural network:

1. **CAV Training:** Given positive concept images P_C and random negative images N_C, extract bottleneck activations $a_l(x) \in \mathbb{R}^{d_l}$. Train a linear classifier; the vector normal to the hyperplane is the Concept Activation Vector $\mathbf{v}_C^l$.

2. **Directional Derivative:** For class k and input x, compute directional derivative along $\mathbf{v}_C^l$:

$$S_{C,k,l}(x) = \nabla_{a_l} h_k(a_l(x)) \cdot \mathbf{v}_C^l \tag{7}$$

3. **TCAV Score:** Calculate the fraction of class k instances with positive directional sensitivity:

$$\text{TCAV}_{Q,k,l} = \frac{|\{x \in X_k : S_{C,k,l}(x) > 0\}|}{|X_k|} \tag{8}$$

2. Systematic Comparison: CBM vs. TCAV

Dimension	Concept Bottleneck Models (CBMs)	Testing with CAVs (TCAV)
Explainability Paradigm	**Explainable-by-Design** (In-Modeling)	**Post-Hoc** (Evaluates pre-trained models)
Model Architecture	Constrained intermediate layer: $x \to c \to y$	Arbitrary standard black-box neural network
Concept Supervision	**Dense annotations**: Every training instance requires ground-truth vector $c^* \in \{0,1\}^K$	**Lightweight exemplar sets**: Requires only 30–50 positive concept images + random negatives
Model Accuracy	Suffers from accuracy-interpretability penalty due to informational bottleneck	Preserves 100% of original black-box performance (no architectural constraint)
Faithfulness	**Faithful by design**: Predictions depend exclusively on predicted concepts	**Approximated**: Linear directional sensitivity at layer l; subject to gradient noise
Human Intervention	**Yes**: Real-time test-time concept intervention ($c_j \leftarrow c_j^*$)	**No**: Strictly passive diagnostic inspection; cannot alter model decisions
Vulnerabilities	Concept leakage, information shortcutting, incomplete concept dictionaries	Layer selection dependence, sensitivity to random negative sets

3. Scenario-Based Preference and Operational Trade-Offs

- **When to Prefer CBMs:** Deploy CBMs in **high-stakes, safety-critical domains where models are designed from scratch** (e.g., medical diagnosis, autonomous vehicle perception). In these domains:
  a. Regulators require by-design architectural guarantees that the model cannot bypass symbolic reasoning.
  b. Human domain experts (doctors, pilots) must have the authority to correct misperceived concepts at runtime to prevent catastrophic errors.
  c. Budget exists to collect comprehensive concept annotations.
- **When to Prefer TCAV:** Deploy TCAV in **auditing and debugging scenarios on pre-trained, proprietary, or foundation models** (e.g., ResNet, ViT, or multimodal LLMs). In these scenarios:
  a. Retraining the architecture is technically impossible or financially prohibitive.
  b. Dense concept labels are unavailable across the primary dataset, but a few dozen concept examples can be easily gathered.
  c. The primary goal is auditing algorithmic bias (e.g., verifying whether gender concepts influence credit/hiring classifications).

★ Examiner Feedback & Common Pitfalls (Prof. Eliana Pastor)

In official grading, students scored full marks (3.00 / 3.00) by correctly balancing both sides: (1) defining both pipelines mathematically, (2) detailing the accuracy vs. faithfulness trade-off, and (3) clearly delineating when to choose by-design intervention versus post-hoc auditing.

➥ **Course Theory Mapping (Notes & Slides)**

NOTES/XAI_Theory.pdf – Chapter 7: "Concept-based Explainable AI", Sections: "Testing with Concept Activation Vectors (T-CAV)" (pp. 92–98) and "Concept Bottleneck Models (CBMs)" (pp. 98–106).

¶ **Alternative Perspectives & Methodological Variants**

Concept Embedding Models (CEMs) bridge the CBM-TCAV trade-off by representing concepts as continuous embeddings $\mathbf{e}_c \in \mathbb{R}^d$, recovering black-box accuracy while retaining test-time concept intervention capabilities.

Question 5: Adversarial Attacks: Danger Motivations and Defense Dilemma — Open Inquiry — 2.50 Points

Describe at least 3 motivations for which Adversarial Attacks are dangerous for a safe and trustworthy deployment of deep learning models.

✓ **Executive Solution Summary**

Adversarial attacks pose severe, fundamental dangers to trustworthy AI deployment due to four critical motivations:

1. **Cross-Model Transferability:** Adversarial perturbations generated against an accessible white-box surrogate model consistently transfer and fool completely different, unseen black-box production models.
2. **Real-World Physical Feasibility:** Using Expectation Over Transformation (EOT), perturbations materialize as physical-world stickers, patches, or 3D objects (e.g., road sign tampering, 3D turtle misclassified as a rifle).
3. **Extension Beyond Vision (LLM Jailbreaking):** Attacks compromise text, audio, and multimodal foundation models, using universal adversarial suffixes (GCG) to strip safety alignment and guardrails.
4. **Crucially, the Complete Absence of Effective/Universal Defenses:** The security arms race leaves deep learning undefended against adaptive attacks; heuristic defenses (defensive distillation) suffer from gradient masking, and adversarial training incurs massive computational overhead while degrading clean accuracy.

✍ **Comprehensive Technical Walkthrough**

1. Mathematical Formulation

An adversarial attack introduces an imperceptible perturbation δ to an input x ($x_{\text{adv}} = x + \delta$ with $\|\delta\|_p \leq \epsilon$) such that $f(x_{\text{adv}}) \neq f(x)$. This exploits the linear nature of deep networks in high-dimensional space (Goodfellow et al., 2014) and the reliance on non-robust features (Ilyas et al., 2019).

2. Four Core Motivations for Adversarial Vulnerability Danger

Motivation 1: Cross-Model Transferability (Black-Box Exploits)

An attacker does not require access to model weights, training data, or architecture. An adversary trains an open-source white-box surrogate model $\hat{f}$, computes gradient-based perturbations (e.g., PGD, C&W), and deploys them against a proprietary black-box system f. Because deep neural networks learn similar decision manifolds for the same task, **adversarial perturbations transfer across distinct architectures, optimizers, and datasets**, rendering proprietary security-by-obscurity completely ineffective.

Motivation 2: Physical-World Realizability (Surviving Reality via EOT)

Adversarial vulnerabilities are not digital artifacts confined to synthetic pixel matrices. Using **Expectation Over Transformation (EOT)** (Athalye et al., 2018):

$$\max_{\delta} \mathbb{E}_{t\sim\mathcal{T}} \left[\mathcal{L}(f(t(x+\delta)), y)\right] \tag{9}$$

adversaries craft physical perturbations that withstand environmental noise, viewing angles, distance, lighting, and printing artifacts:

- *Autonomous Driving Exploits:* Strategic stickers placed on a STOP sign cause YOLO and MobileNet vision models to classify it as an "80 km/h Speed Limit" sign, causing intersections to be crossed at high speed.
- *3D Printed Physical Attacks:* A physical 3D-printed plastic turtle textured with an adversarial pattern is classified by Google Inception-v3 as a "Rifle" from virtually every camera angle.
- *Surveillance Evasion:* Fabric printed with adversarial camouflage patterns (Adv-Shirt) disables bounding-box detectors, rendering armed individuals invisible to automated airport surveillance.

Motivation 3: Extension Beyond Vision to Foundation Models and LLM Jailbreaks

Adversarial vulnerability is universal across all machine learning modalities:

- *Large Language Models (LLMs):* Automated token optimization (Greedy Coordinate Gradient – GCG; Zou et al., 2023) generates universal adversarial suffix strings (e.g., sequences of symbols appended to malicious prompts). These suffixes completely bypass Reinforcement Learning from Human Feedback (RLHF), safety filters, and system prompt guardrails, forcing LLMs to output actionable instructions for cyberattacks or dangerous chemical synthesis.
- *Speech and Audio Systems:* Inaudible acoustic noise embedded in background music reliably triggers smart home voice assistants to unlock doors or visit malicious URLs.

Motivation 4: The Absence of Universal or Effective Defenses (The Arms Race)

The single most alarming danger in trustworthy AI is that **there exists no general, reliable defense against adversarial attacks**:

- **Failure of Heuristic Defenses (Gradient Masking):** Early defense strategies, such as *Defensive Distillation* (Papernot et al.), merely smoothed or flattened the model's output probabilities, driving gradients to near-zero ($\nabla_x \mathcal{L} \approx 0$). While this broke simple one-step attacks (FGSM), Carlini & Wagner (2017) demonstrated that gradient masking provides false security: optimization-based attacks and surrogate models bypass it effortlessly.

- **Adversarial Training Trade-Offs and Overhead:** The only currently accepted empirical defense is **Adversarial Training**:

$$\min_{\theta} \mathbb{E}_{(x,y)} \left[\max_{\|\delta\|_p \leq \epsilon} \mathcal{L}(f_\theta(x+\delta), y) \right] \tag{10}$$

  However, adversarial training: (1) requires massive computational overhead (multiplying training time by $5\times$ to $10\times$ due to inner-loop PGD optimization), (2) induces a **robustness-accuracy trade-off** (clean test accuracy noticeably drops), and (3) is strictly norm-specific (a model trained to resist L_∞ attacks remains utterly defenseless against L_2 or L_0 attacks).

★ Examiner Feedback & Common Pitfalls (Prof. Eliana Pastor)

In official exam feedback, Prof. Eliana Pastor docked -0.25 points when candidates discussed transferability, LLM jailbreaks, and physical attacks, but **failed to state the absence of effective/universal defenses**. Mentioning the defense dilemma, defensive distillation flaws, and the arms race guarantees full marks (2.50 / 2.50).

➥ Course Theory Mapping (Notes & Slides)

NOTES/XAI_Theory.pdf – Chapter 12: "Adversarial Attacks and Defenses", Sections: "Why are Adversarial Attacks Important?", "Is the Threat Real?", "Transferability", and "Adversarial Defenses" (pp. 175–189).

¶ Alternative Perspectives & Methodological Variants

Certified Robustness (Randomized Smoothing) mathematically guarantees that the classification of x remains constant within an L_2 radius $R = \sigma\Phi^{-1}(p_A)$, providing verifiable safety bounds rather than empirical heuristics.

Question 6: Evaluating Saliency Maps in Clinical Tumor Classification

Open Inquiry | 2.50 Points

A hospital uses saliency maps to explain an image classifier for detecting tumors. The maps often highlight clinically plausible regions, but the hospital is concerned that the explanations may not reflect the actual model reasoning.
Design an evaluation strategy for these explanations. Discuss faithfulness, plausibility, and at least two evaluation approaches.

✓ Executive Solution Summary

In clinical tumor diagnosis, an explanation can be highly **plausible** (highlighting anatomical areas oncologists intuitively expect) while being entirely **unfaithful** (concealing that the neural network actually made its prediction based on a metallic hospital token, patient positioning, or scanner artifacts).
We design a rigorous evaluation strategy by establishing: (1) formal definitions of faithfulness (correctness/necessity and completeness/sufficiency) versus plausibility, and (2) two quantitative evaluation approaches: **Removal-Based Evaluation (AOPC / Feature Erasure curves)** and **Parameter and Data Randomization Sanity Checks** (Adebayo et al.).

✍ **Comprehensive Technical Walkthrough**

1. Faithfulness versus Plausibility in Clinical Diagnostics

- **Plausibility (Human Alignment / Convincingness):** Measures whether the highlighted regions correspond to human medical knowledge (e.g., overlapping with radiologist-annotated tumor margins). Plausibility can be quantified using the Dice coefficient or Intersection over Union (IoU) between the saliency map $M(x)$ and clinical ground-truth segmentations $S_{\text{doctor}}(x)$:

$$\text{IoU} = \frac{|M(x) \cap S_{\text{doctor}}(x)|}{|M(x) \cup S_{\text{doctor}}(x)|} \tag{11}$$

  Crucial Clinical Risk: A high IoU does not prove the model reasoned clinically. A model relying on hospital watermarks can generate a plausible-looking explanation by accident or through edge-detection artifacts.

- **Faithfulness (Fidelity / Truthfulness):** Measures how accurately the explanation reflects the model's actual internal computational mechanism. Faithfulness is decomposed into two foundational properties:

  a. **Correctness / Necessity:** Do the highlighted pixels actually contribute to the prediction? Removing them must significantly drop model confidence.

  b. **Completeness / Sufficiency:** Are the highlighted pixels sufficient on their own to retain the prediction without the rest of the image?

2. Two Quantitative Evaluation Approaches

Approach 1: Removal-Based Evaluation (Feature Erasure & AOPC Curves)

To test faithfulness without human subjectivity, we measure the causal impact on model confidence $f_c(x)$ when pixels are systematically removed (masked with mean dataset color, blurred, or infilled via generative inpainting):

Erasure Strategy	Algorithmic Execution	Expected Behavior for Faithful Explanation
Most-Salient-First (MoSF) [**Necessity**]	Sort pixels by descending attribution; iteratively delete top $k\%$ most salient features	**Steep, immediate drop** in tumor class probability $f_c(x)$ as crucial evidence is destroyed.
Least-Salient-First (LeSF) [**Sufficiency**]	Sort pixels by ascending attribution; iteratively delete bottom $k\%$ least salient features	**Flat, invariant confidence** curve; prediction must remain stable until necessary features are touched.

Quantitative Metric — Area Over the Perturbation Curve (AOPC):

$$\text{AOPC}_K = \frac{1}{K} \sum_{k=1}^{K} \left(f_c(x^{(0)}) - f_c(x^{(k)}) \right) \tag{12}$$

where $x^{(k)}$ is the image after k steps of MoSF erasure. A larger AOPC indicates superior necessity and faithfulness. If removing the top salient pixels causes no drop in tumor score, the saliency map is provably unfaithful.

Approach 2: Model Parameter and Data Randomization Sanity Checks (Adebayo et al., 2018)

Many popular saliency methods (e.g., Guided Backpropagation, Guided Grad-CAM) act as generic image edge detectors rather than explanation tools. We perform the **Cascading Randomization Sanity Check**:

Step 1. Generate baseline saliency map $M_{\text{trained}}(x)$ using the trained diagnostic model.

Step 2. Systematically randomize the model weights layer-by-layer, beginning from the top classification layer down to the earliest convolutional layer:

$$W^{(l)} \sim \mathcal{N}\left(0, \sigma^2\right) \quad \text{for } l = L, L-1, \ldots, 1 \tag{13}$$

Step 3. Recompute the saliency map $M^{(l)}_{\text{random}}(x)$ at each step and calculate structural similarity (SSIM) or Pearson correlation with the original map:

$$\rho = \text{Corr}\left(M_{\text{trained}}(x), M^{(l)}_{\text{random}}(x)\right) \tag{14}$$

Step 4. **Evaluation Criterion:** A faithful explanation method **must degrade and collapse** as weights are randomized ($\rho \to 0$). If the saliency map remains crisp and identical even when all model parameters are completely random, the method is *implementation-invariant*, acts as a trivial edge detector, and is **100% unfaithful to the hospital model**.

★ Examiner Feedback & Common Pitfalls (Prof. Eliana Pastor)

Ensure you define both sub-properties of faithfulness: **correctness/necessity** (tested via Most-Salient-First deletion) and **completeness/sufficiency** (tested via Least-Salient-First deletion). For sanity checks, explicitly name the cascading parameter randomization test developed by Adebayo et al.

➥ Course Theory Mapping (Notes & Slides)

NOTES/XAI_Theory.pdf – Chapter 6: "Evaluation of Explanations", Sections: "Plausibility vs. Faithfulness" (pp. 82–86) and "Evaluating Content / Model Properties: Input Erasure and Sanity Checks" (pp. 86–91).

¶ Alternative Perspectives & Methodological Variants

The Infidelity Metric (Yeh et al., 2019) mathematically formalizes faithfulness by computing the expected mean squared error between the inner product of the perturbation vector with the explanation and the actual change in model output: $\mathbb{E}_{I \sim \mu_I}[(I^T \Phi(f, x) - (f(x) - f(x - I)))^2]$.

Question 7: Tabular Student Dropout Prediction: XAI Design and Dual-Stakeholder Recommendation

Open Inquiry | 4.50 Points

A university uses an AI system to predict whether students are at high risk of dropping out. The model operates on tabular data, such as exam results, number of credits obtained, attendance, previous failures, and demographic information. For each student, the system outputs a binary

prediction indicating whether the student is at high risk, together with the corresponding prediction probability.
The university wants to provide explanations that are useful both for academic advisors and for students. Discuss at least three explainability methods that could be used to explain the behavior or predictions of the risk model. For each method, specify whether it provides a local or global explanation and describe the type of explanation representation it produces. Then, choose one method you would recommend for this scenario and justify your choice.

✓ Executive Solution Summary

Predicting student dropout from tabular data involves continuous variables (GPA, credits), discrete counts (failures), and demographic attributes. The system serves two distinct stakeholders: **Academic Advisors** (who require global insights into systemic educational bottlenecks and local risk audits) and **Students** (who require actionable recourse without being discouraged by non-modifiable demographic factors).
We evaluate four methods: (1) **TreeSHAP/SHAP** (Local & Global, Additive feature attributions), (2) **Actionable Counterfactuals / DiCE** (Local, Example-based actionable recourse), (3) **Permutation Feature Importance** (Global, Feature importance rankings), and (4) **LIME** (Local, Sparse linear surrogate).
We recommend deploying a unified framework centered on **TreeSHAP paired with Actionable Counterfactual Recourse (DiCE)** (with SHAP as primary): SHAP empowers academic advisors with cohort-wide beeswarm distributions and individual attribution waterfalls, while DiCE supplies students with concrete, feasible study milestones while strictly freezing immutable demographic features.

✍ **Comprehensive Technical Walkthrough**

1. Operational Context & Tabular Data Characteristics

The university AI system processes structured tabular student records:

- *Academic Continuous Features:* Number of credits obtained (ECTS), average grade (GPA), lecture attendance percentage ($[0, 100]\%$).
- *Academic Discrete Features:* Number of past exam failures, semester count.
- *Demographic / Socioeconomic Features:* Age, gender, commuter status, family income bracket, parental education level.

Dual-Stakeholder Profile and Requirements

Stakeholder 1. **Academic Advisors:** Need both global and local perspectives. Globally, they must identify systemic failure trends across student cohorts (e.g., identifying gateway courses causing widespread failures) to adjust curriculum schedules. Locally, they need a clear breakdown during 1-on-1 advisory meetings to understand why a student is flagged as high risk.

Stakeholder 2. **Students:** Need constructive, actionable, and non-discriminatory guidance. Telling a student *"You are high risk because of your socio-economic status and age"* is legally unacceptable, non-actionable, and psychologically damaging. Students need **actionable recourse**: *"What concrete academic goals must I achieve this semester to exit high-risk status?"*

2. Analysis of Candidate Explainability Methods

Method 1: SHAP (TreeSHAP) [Local & Global | Feature Attribution]

- **Scope:** Both strictly **local** (explaining individual student predictions via Shapley values) and **global** (aggregating local attributions across all students).
- **Mechanism:** Computes the fair marginal contribution ϕ_j of each feature across all possible feature coalitions, satisfying the Efficiency axiom:

$$f(x) = \mathbb{E}[f(X)] + \sum_{j=1}^{M} \phi_j \tag{15}$$

  In gradient-boosted trees (XGBoost, LightGBM), TreeSHAP calculates exact attributions in polynomial time $\mathcal{O}(TLD^2)$.
- **Explanation Representation:**
  - *Local Representation:* Waterfall and Force plots showing signed contributions (e.g., $\phi_{\text{Credits}} = +0.32$, $\phi_{\text{Attendance}} = +0.21$, $\phi_{\text{Failures}} = +0.18$, $\phi_{\text{Age}} = +0.02$).
  - *Global Representation:* Beeswarm summary plots and Dependence plots displaying feature distributions and non-linear threshold effects across the entire university population.

Method 2: Counterfactual Recourse (DiCE) [Local | Example-Based Recourse]

- **Scope:** Strictly **local**.
- **Mechanism:** Generates a set of diverse counterfactual instances x^* that reverse the risk classification from Risk = 1 to Risk = 0, solving:

$$x^* = \arg\min_{x'} d_{\text{MAD}}(x, x') + \lambda \left(f(x') - y_{\text{low_risk}}\right)^2 \quad \text{s.t.} \quad x'_j = x_j \; \forall j \in \mathcal{X}_{\text{immutable}} \tag{16}$$

  where distance is normalized by the Median Absolute Deviation (d_{MAD}).
- **Constraint Handling:** Features are explicitly constrained:
  - *Immutable Features:* $\mathcal{X}_{\text{immutable}} = \{\text{Age}, \text{Gender}, \text{Commuter}, \text{Socioeconomic Status}\}$ are strictly frozen ($\Delta x_j = 0$).
  - *Actionable Features:* $\mathcal{X}_{\text{actionable}} = \{\text{Credits}, \text{Attendance}, \text{Exam Registrations}\}$ are allowed to vary in feasible directions ($\Delta\text{Credits} \geq 0$).
- **Explanation Representation:** Example-based comparison: *"You are currently high risk ($P = 0.78$). You will become low risk ($P = 0.22$) if you: (1) Earn ≥ 12 credits this session, and (2) Increase attendance to $\geq 80\%$."*

Method 3: Permutation Feature Importance (PFI) [Global | Feature Importance]

- **Scope:** Strictly **global**.
- **Mechanism:** Evaluates the drop in model accuracy after randomly permuting feature column j across the dataset: $I(j) = \mathcal{L}(y, f(\tilde{X}^{(j)})) - \mathcal{L}(y, f(X))$.
- **Explanation Representation:** Bar chart ranking global factors (e.g., Credits > Attendance > Failures).
- **Limitation:** Produces no local explanations for individual students.

Method 4: LIME [Local | Sparse Linear Attribution]

- **Scope:** Strictly **local**.
- **Mechanism:** Fits a local weighted linear model on perturbed student profiles.
- **Limitation on Tabular Data:** Highly sensitive to kernel bandwidth σ; suffers from sampling instability where consecutive runs yield conflicting weights for the same student.

3. Comparative Synthesis Across Stakeholder Needs

Method	Scope	Representation	Value for Advisors	Value for Students
SHAP	Local & Global	Additive feature attribution	**Optimal**: Cohort risk trends + student audit	High understanding of why risk was assigned
DiCE	Local	Actionable counterfactual example	Validates intervention feasibility	**Optimal**: Actionable study roadmap
PFI	Global	Dataset importance ranking	Broad overview of factors	Zero value (no individual guidance)
LIME	Local	Linear surrogate weights	Vulnerable to sample variance	Confusing if weights fluctuate

4. Final Recommendation and Rigorous Justification

Recommendation: We formally recommend implementing **SHAP (TreeSHAP)** as the core foundational engine, integrated with an **Actionable Counterfactual Recourse module (DiCE)** on the student portal.

Dual-Stakeholder Justification:

J1. For Academic Advisors (Auditability, Efficiency, and Cohort Strategy):

- *Global Strategy:* Advisors use SHAP beeswarm summary plots and dependence plots to discover systemic institutional bottlenecks (e.g., recognizing that earning fewer than 18 credits in Semester 1 exponentially accelerates dropout risk across all majors).
- *Local Diagnostic Audit:* When meeting an at-risk student, advisors inspect the SHAP waterfall plot to understand the precise combination of academic indicators driving the score, backed by game-theoretic axiomatic fairness.

J2. For Students (Constructive Actionability & Ethical Non-Discrimination):

- *Actionable Roadmap over Passive Guilt:* Traditional feature attribution merely informs students of their deficits. By integrating counterfactual recourse, the system translates mathematical predictions into a constructive study plan.
- *Ethical and Regulatory Constraints:* Counterfactual generation strictly freezes immutable demographic variables ($\Delta x_{\text{demographic}} = 0$). This guarantees that explanations never give discriminatory or offensive recommendations (such as implying a student is at risk due to age, gender, or background), ensuring strict alignment with the EU AI Act and GDPR Article 22 mandates.

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This question earned full marks (4.50 / 4.50) in official evaluations. Key grading factors:

- Clearly distinguishing between local and global scopes for each method.
- Addressing the tabular domain characteristics (continuous vs. categorical variables).
- Providing a dual-stakeholder justification: distinguishing advisor needs (cohort trends, diagnostic audit) from student needs (actionability, non-discriminatory recourse).
- Explicitly discussing immutable demographic features and why they must be constrained in counterfactual recourse.

➥ **Course Theory Mapping (Notes & Slides)**

NOTES/XAI_Theory.pdf – Chapter 5: "Post-Modeling Explainability: SHAP" (pp. 54–62) and Chapter 9: "Counterfactual Explanations: DiCE and Actionability" (pp. 130–138).

¶ **Alternative Perspectives & Methodological Variants**

Prescriptive Trees (Bertsimas et al.) train optimal decision trees that directly map student profiles to optimal intervention actions (e.g., assigning a tutoring mentor or granting a scholarship) rather than simply predicting risk scores.